

Normalized spectral-ball automorphisms are global holomorphic conjugations

Candidate solution packet for arXiv:1011.2485

13 August 2026

Status. Candidate full solution, likely valid, human review needed.

1 Source question and scope

Let

$$\Omega_n = \{X \in \mathcal{M}_n(\mathbb{C}) : \rho(X) < 1\}$$

be the spectral ball. In the final paragraph of Kosiński's note [1] (PDF page 3), the first of two problems asks:

Is every normalized automorphism of the spectral ball of the form $X \mapsto u(X)Xu(X)^{-1}$, where $u : \Omega_n \rightarrow \mathrm{GL}_n(\mathbb{C})$ is holomorphic?

Here normalized means $F(0) = 0$ and $F'(0) = \mathrm{id}$. The source evidence is reproduced below.

Remark 2. We would like to point out that the situation occurring in the above example may be described more precisely. Namely, put $u := \begin{pmatrix} 1 & 0 \\ a & 1 \end{pmatrix}$, where a is an arbitrary holomorphic function on Ω . Then the conjugation $\mathcal{J}_u : x \mapsto u(x)xu(x)^{-1}$ is an automorphism of Ω if and only if a depends on x_{12} , $\text{tr } x$ and $\det x$.

The problem of describing the group of automorphisms of spectral ball was divided by Rostand ([4]) into two parts. The first one is to prove that any normalized automorphism (i.e. an automorphism fixing the origin and satisfying $f'(0) = \text{id}$) is a conjugation $\mathcal{J}_u$, where $u : \Omega \rightarrow \mathcal{M}_n^{-1}$ is a holomorphic mapping. The second one is to show that the automorphism being a conjugation is of the form $\mathcal{J}_u$ with u satisfying the condition $u(q^{-1}xq) = u(x)$ for each $x \in \Omega$ and each invertible q . Note that it was shown in this paper that the second statement is false. On the other hand it seems that the answer to the first question is affirmative. So we have two following problems:

- Is every normalized automorphism of the spectral ball of the form $x \mapsto u(x)xu(x)^{-1}$, where $u : \Omega \rightarrow \mathcal{M}_n^{-1}$ is a holomorphic mapping?
- Describe all holomorphic $u : \Omega \rightarrow \mathcal{M}_n^{-1}$ such that the mapping $x \mapsto u(x)xu(x)^{-1}$ is an automorphism of Ω .

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INSTITUTE OF MATHEMATICS, JAGIELLONIAN UNIVERSITY, ŁOJASIEWICZA 6, 30-348 KRAKÓW, POLAND
E-mail address: lukasz.kosinski@gazeta.pl

The theorem below answers this first problem affirmatively for every n . It does not classify all holomorphic u whose conjugation map is an automorphism, which is the source's separate second problem.

2 Proof intuition

On the open set of cyclic matrices, conjugators exist locally and any two differ by an invertible matrix commuting with X . The commuting algebra of a cyclic matrix is exactly the polynomial algebra generated by X . Globally, this is the function algebra of the spectral cover

$$Y = \{(X, \lambda) : \det(\lambda I - X) = 0\} \longrightarrow \Omega_n.$$

Thus the failure of the local conjugators to glue is measured by a line bundle on the part of Y lying over cyclic matrices.

Two ingredients kill that obstruction. First, the normalized rescaling $F_t(X) = t^{-1}F(tX)$ deforms F through spectrum-preserving automorphisms to the identity; local conjugators chosen by Krylov bases make the obstruction line bundle vary continuously, so its first Chern class is zero. Second, the deleted part of the spectral cover has codimension three. A depth/local cohomology argument gives $H^1(\mathcal{O}) = 0$ on its complement, making the Chern class map on the Picard group injective. The obstruction is therefore holomorphically trivial. After gluing on cyclic matrices, ordinary Hartogs extension crosses the codimension-three derogatory locus; purity of the zero set of a holomorphic function ensures that the extended conjugator remains invertible.

3 Geometric lemmas

Call X *cyclic* if some $v \in \mathbb{C}^n$ makes $v, Xv, \dots, X^{n-1}v$ a basis, and let $D \subset \Omega_n$ be the derogatory (noncyclic) locus. Put $U = \Omega_n \setminus D$.

Lemma 1 (Codimension and the spectral cover). *For $n \geq 2$, D is analytic of codimension three. The finite flat spectral cover*

$$\pi : Y := \{(X, \lambda) \in \Omega_n \times \mathbb{C} : \det(\lambda I - X) = 0\} \longrightarrow \Omega_n$$

has degree n . It is a Stein Cohen–Macaulay, normal hypersurface. If $W = \pi^{-1}(D)$ and $V = Y \setminus W = \pi^{-1}(U)$, then

$$\mathrm{codim}_Y W = 3, \quad H^1(V, \mathcal{O}_V) = 0.$$

Proof. A matrix is derogatory precisely when some eigenvalue has geometric multiplicity at least two. Hence D is the projection of

$$S = \{(X, \lambda) : \mathrm{rank}(X - \lambda I) \leq n - 2\}.$$

For fixed λ , the rank condition has codimension four in $\mathcal{M}_n$. Consequently S has dimension $n^2 - 3$ in $\mathcal{M}_n \times \mathbb{C}$. Its projection has finite fibers, so D has dimension $n^2 - 3$ (the generic stratum with one diagonalizable double eigenvalue realizes this dimension).

The equation defining Y is monic of degree n in λ , so π is finite flat of degree n . The spectral ball is pseudoconvex and therefore Stein; hence Y , a closed analytic hypersurface in $\Omega_n \times \mathbb{C}$, is Stein. Every hypersurface is Cohen–Macaulay. Differentiating the determinant shows that a point of Y is singular exactly when $\mathrm{rank}(\lambda I - X) \leq n - 2$: the matrix derivatives are the entries of the adjugate. Thus the singular locus has codimension three in Y , and Serre’s hypersurface criterion implies that Y is normal.

Because π is finite and D has codimension three, so does $W = \pi^{-1}(D)$. The analytic Cohen–Macaulay depth theorem [5] gives

$$H_W^q(Y, \mathcal{O}_Y) = 0 \quad (q < 3).$$

Cartan’s theorem B [6] gives $H^q(Y, \mathcal{O}_Y) = 0$ for $q > 0$. The long exact local-cohomology sequence for (Y, W) now yields

$$H^1(V, \mathcal{O}_V) = 0,$$

as asserted. Notice that for $n > 2$, W can strictly contain $\text{Sing } Y$; only its codimension and the Cohen–Macaulay depth are used. $\square$

Lemma 2 (Centralizers and the obstruction bundle). *Let $G : U \rightarrow \mathcal{M}_n$ be holomorphic, spectrum-preserving, and pointwise conjugate to the identity map. Local holomorphic conjugators $G(X) = p_i(X)Xp_i(X)^{-1}$ determine a holomorphic line bundle L_G on V . The bundle is trivial if and only if there is one holomorphic $p : U \rightarrow \text{GL}_n(\mathbb{C})$ satisfying $G(X) = p(X)Xp(X)^{-1}$.*

Proof. Local conjugators exist directly. At $X_0 \in U$, choose a vector v cyclic for both X_0 and $G(X_0)$; such a vector exists because each set of cyclic vectors is a nonempty Zariski-open subset of $\mathbb{C}^n$. After shrinking, the Krylov matrices

$$K_v(A) = [v, Av, \dots, A^{n-1}v]$$

are invertible for $A = X$ and $A = G(X)$. Since X and $G(X)$ have the same characteristic polynomial,

$$p(X) = K_v(G(X))K_v(X)^{-1}$$

satisfies $G(X) = p(X)Xp(X)^{-1}$.

On an overlap, $c_{ij} = p_i^{-1}p_j$ commutes with X . For cyclic X , its centralizer is $\mathbb{C}[X]$. In sheaf form this gives the canonical algebra isomorphism

$$\pi_*\mathcal{O}_V \cong \mathcal{O}_U[t]/(\det(tI - X)) \cong \{A \in \mathcal{M}_n(\mathcal{O}_U) : AX = XA\}.$$

Thus every c_{ij} is a unit in $\pi_*\mathcal{O}_V$, equivalently a nowhere-zero holomorphic transition function on V . The cocycle relation defines a line bundle L_G .

If L_G is trivial, write $c_{ij} = h_i^{-1}h_j$ with h_i units of $\pi_*\mathcal{O}_V$, hence invertible centralizer matrices. Then $p_i h_i^{-1} = p_j h_j^{-1}$ on overlaps, and these matrices glue to the desired global conjugator. The converse follows by comparing any local conjugators with a global one. $\square$

4 Main theorem

Theorem 3. *For every $n \geq 1$, every normalized holomorphic automorphism $F \in \text{Aut}(\Omega_n)$ admits a holomorphic map $u : \Omega_n \rightarrow \text{GL}_n(\mathbb{C})$ such that*

$$F(X) = u(X)Xu(X)^{-1} \quad (X \in \Omega_n).$$

Proof. The case $n = 1$ is immediate. Assume $n \geq 2$. A normalized automorphism of the spectral ball preserves the spectrum [2]; in particular, on U it sends every cyclic X to a cyclic matrix conjugate to X .

For $0 < t \leq 1$ define

$$F_t(X) = t^{-1}F(tX), \quad F_0(X) = X.$$

The Taylor expansion at zero and $F'(0) = \text{id}$ show continuity at $t = 0$. Spectrum preservation gives $\text{Spec } F_t(X) = \text{Spec } X$. Moreover $t^{-1}F^{-1}(tX)$ is the inverse of F_t , so every F_t is an automorphism.

Apply Lemma 2 to $F_t|_U$. The local Krylov construction in its proof is parametric: near any $(t_0, X_0) \in [0, 1] \times U$, choose one vector cyclic for both X_0 and $F_{t_0}(X_0)$; invertibility persists nearby. The resulting transition functions therefore define a continuous complex line bundle $\mathcal{L}$ on $[0, 1] \times V$, whose restriction at t is the underlying topological bundle of L_{F_t} . At $t = 0$, the Krylov formula gives the identity conjugator, so $\mathcal{L}|_{\{0\} \times V}$ is trivial. Homotopy invariance now gives

$$c_1(L_F) = c_1(L_{F_1}) = 0 \quad \text{in } H^2(V, \mathbb{Z}).$$

The exponential sequence on V contains

$$H^1(V, \mathcal{O}_V) \longrightarrow \text{Pic}(V) \xrightarrow{c_1} H^2(V, \mathbb{Z}).$$

Lemma 1 makes the left group zero, so c_1 is injective. It follows that L_F is holomorphically trivial. Lemma 2 therefore supplies a holomorphic $u : U \rightarrow \mathrm{GL}_n(\mathbb{C})$ with $F(X) = u(X)Xu(X)^{-1}$.

Finally, $D = \Omega_n \setminus U$ has codimension three. Hartogs extension [6] extends every matrix entry of u holomorphically across D , giving $\tilde{u} : \Omega_n \rightarrow \mathcal{M}_n$. Its determinant is nonzero on U . If $\det \tilde{u}$ vanished, its zero set would be a divisor, but it would have to lie inside the codimension-three set D , which is impossible. Therefore $\tilde{u}$ takes values in $\mathrm{GL}_n(\mathbb{C})$, and the conjugation identity extends to all of Ω_n by holomorphic continuity. $\square$

5 Verification, limitations, and novelty

The proof was stress-tested in all dimensions, at ramified cyclic fibers, and at the derogatory locus. The distinction $W = \pi^{-1}(D) \supseteq \mathrm{Sing} Y$ for $n > 2$ is explicit; the depth argument applies to W itself. No computation is used as proof.

A bounded literature search on 13 August 2026 checked exact and close variants of “spectral ball global holomorphic conjugation”, the source [1], Thomas’s local theorem [3], and Kosiński’s later $n = 2$ local/density theorem [4]. The latter states that automorphisms of Ω_2 are *locally* holomorphic conjugations and that certain conjugations generate a dense subgroup; it does not state the global theorem above. No exact prior answer was found. Novelty confidence is moderate because the search was bounded.

The source’s second final problem, a classification of all holomorphic u whose conjugation map is an automorphism, remains outside this packet. Human review should concentrate on the local-cohomology vanishing in Lemma 1 and the parametric line-bundle construction in Theorem 3.

References

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